Activity 7 Pre-Lab — Grignard: Synthesis of Triphenylmethanol

Organic Lab · Grignard reagent + addition to a ketone

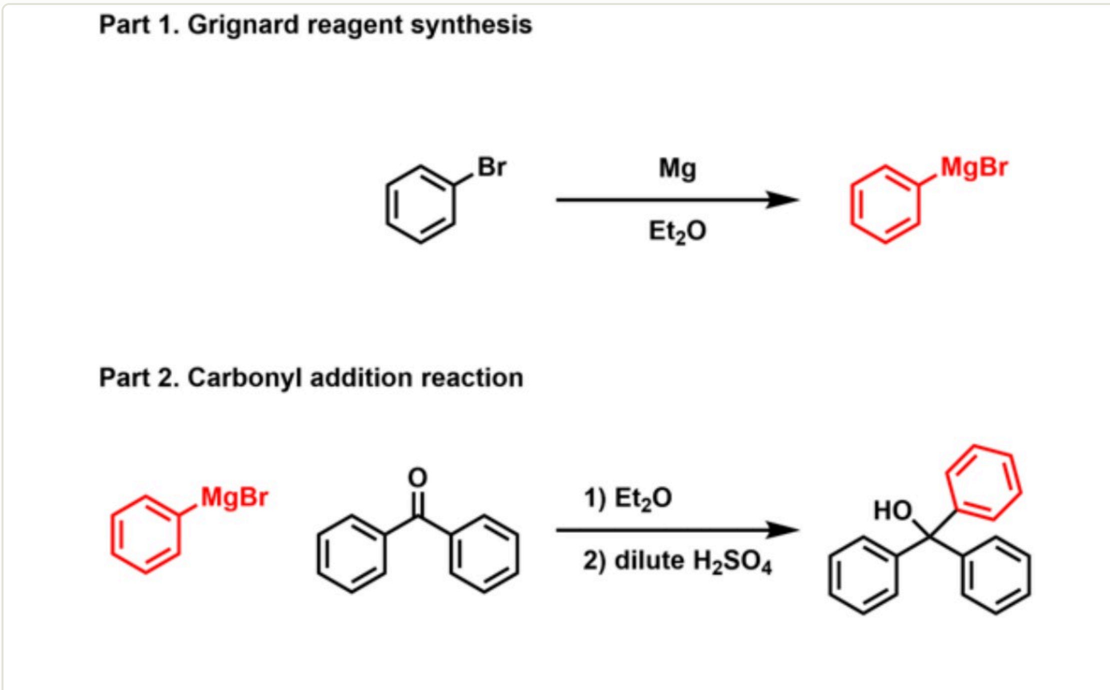
[Your name] & [partner]
[MM/DD/YY]

Black = write this in your notebook **Blue = context, don't copy it** Copy the header onto every page. Draw dashed-box structures yourself.

Purpose

Make the **Grignard reagent phenylmagnesium bromide** from bromobenzene + Mg, then use it as a **carbon nucleophile** that adds to the carbonyl of **benzophenone**. Acidic work-up gives **triphenylmethanol**, purified by recrystallization.

✍ **DRAW — reproduce both steps in your notebook (from the procedure)**



Part 1 makes PhMgBr (dry Et_2O, I_2 initiator); Part 2: PhMgBr adds to benzophenone, then dilute H_2SO_4 work-up protonates the alkoxide → triphenylmethanol.

CONTEXT ▸ ALL glassware must be bone-dry — water quenches (destroys) the Grignard. Don't wash glassware the day of lab.

Chemical Reagents & Safety Table

CONTEXT ▸ Required: magnesium, iodine, diethyl ether, bromobenzene, benzophenone, sulfuric acid. Typical SDS values — cross-check the SDS tile.

Chemical	Role	MW	Amount used	Physical properties	GHS pictograms
Magnesium (turnings)	Makes the Grignard	24.31	0.5 g (~0.021 mol)	Silvery metal solid	flammable
Iodine	Initiator (activates Mg)	253.81	1 small crystal	Purple-black solid; sublimes	irritant, health, environ.
Diethyl ether (anhydrous)	Dry solvent	74.12	12 + 5 + 30 mL	Colorless liquid; bp 34.6 °C; d 0.71 g/mL; flash pt –45 °C	flammable, irritant
Bromobenzene	Grignard precursor (limiting)	157.01	2.0 mL (~3.0 g, ~0.019 mol)	Colorless liquid; bp 156 °C; d 1.50 g/mL; flash pt 51 °C	flammable, irritant, environ.
Benzophenone	Electrophile (ketone)	182.22	3.2 g (~0.018 mol) in 30 mL Et_2O	White solid; mp 48 °C	irritant, health, environ.
Sulfuric acid (conc. → dil.)	Acidic work-up (protonates alkoxide)	98.08	a few mL conc. → into 7 mL water	Colorless viscous liquid; d 1.84 g/mL	corrosive

FYI ▸ also used: anhydrous $CaCl_2$ (drying tube), methanol (recrystallization solvent).

GRAB FIRST (all DRY): 250-mL round-bottom · stir bar + stir plate · Claisen adapter · condenser + thermometer adapter · drying tube + cotton + anhydrous $CaCl_2$ · 125-mL dropping funnel · iron rings + clamps · Tygon tubing · steam bath (Part 2) · Büchner + filter flask (Part 3).

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Part 1 — Make PhMgBr

1. **0.5 g Mg turnings** + stir bar in a **dry 250-mL RBF**; stir vigorously 5–10 min (stopper on). Grinding strips the MgO coat.
2. Pack a **drying tube**: cotton plug → anhydrous $CaCl_2$ (not tight) → cotton.
3. Instructor drops in **1 small I_2 crystal** (initiator).
4. Build the **reflux apparatus**: Claisen on the RBF → straight arm gets condenser + thermometer adapter + drying tube; curved arm gets the **125-mL dropping funnel**. Turn condenser water to a steady drip.
5. Add solutions via the funnel **off the apparatus** each time (stopcock closed while filling): first **12 mL anhydrous Et_2O** → run onto the Mg. Note color changes.
6. Next add **2.0 mL bromobenzene + 5 mL Et_2O** dropwise; the mix should self-reflux (cloudy/grey = PhMgBr forming). If it won't start: warm gently / crush Mg with a stir rod.

Part 2 — Add benzophenone

7. Add **3.2 g benzophenone in 30 mL Et_2O** dropwise via the funnel; reflux (steam bath) as directed.

Part 3 — Work-up + purify

8. Prepare cold dilute acid: add conc. H_2SO_4 to **7 mL DI water** (acid to water!). Remove drying tube; add all the cold acid to quench the alkoxide → Ph_3C–OH. Conc. H_2SO_4 is extremely corrosive.
9. Isolate crude solid; **recrystallize from methanol** (see Activity 3). Weigh → **% yield**; take mp / IR.

✍ **WRITE**
limiting reagent = bromobenzene (0.019 mol) → theoretical Ph_3C–OH; % yield = actual ÷ theoretical × 100

Experiment Notes (in lab)

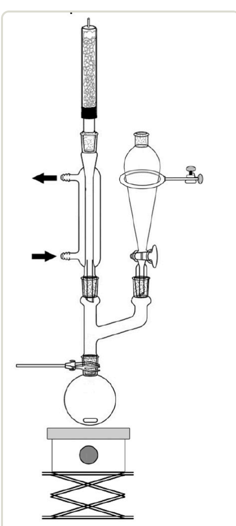
measurement	value
Mass Mg	g
Vol bromobenzene	mL
Crude Ph_3C–OH mass	g
Recrystallized mass	g
mp of product	°C
% yield	

Colors on adding Et_2O / bromobenzene:

Signs Grignard formed (reflux? cloudy grey?):

On adding benzophenone / work-up:

Setup — draw this



reflux apparatus (drying tube · condenser · dropping funnel · Claisen)

CONTEXT (don't copy) ▸

Why dry / drying tube? PhMgBr is a strong base — any water protonates it to benzene and it's dead. The $CaCl_2$ tube keeps humid air out.

Why I_2? a tiny bit of iodine cleans the Mg surface so the reaction starts.

Why add acid last? the first product is an alkoxide salt; H_3O^+ protonates it to the neutral alcohol (triphenylmethanol).

Clean-up & Conclusions

- Ether & organic filtrates → organic waste; acid aqueous → neutralize/aqueous waste; return glassware; bench task.
- **Conclusions** (~4 sentences after lab): % yield + why (moisture killed some Grignard, transfer losses); what mp/IR confirm about purity/identity; why anhydrous conditions mattered.